Classification based on Pharmacological effect:

In this classification, the drugs are grouped based on their biological effect that they produce on the recipient. For example, the medicines that have the ability to kill the pathogenic bacteria are grouped as antibiotics. This kind of grouping will provide the full range of drugs that can be used for a particular condition (disease). The physician has to carefully choose a suitable medicine from the available drugs based on the clinical condition of the recipient.

Examples:

Antibiotic drugs: amoxicillin, ampicillin,cefixime, cefpodoxime, erythromycin, tetracycline etc..

Antihypertensive drugs: propranolol, atenolol, metoprolol succinate, amlodipine etc…

Classification based on the target system (drug action):

In this classification, the drugs are grouped based on the biological system/process, that they target in the recipient. This classification is more specific than the pharmacological classification.For example, the antibiotics streptomycin and erythromycin inhibit the protein synthesis (target process) in bacteria and are classified in a same group. However, their mode of action is different. Streptomycin inhibits the initiation of protein synthesis, while erythromycin prevents the incorporation of new amino acids to the protein.

Classification based on the site of action (molecular target):

The drug molecule interacts with biomolecules such as enzymes, receptors etc,, which are referred as drug targets. We can classify the drug based on the drug target with which it binds. This classification is highly specific compared to the others. These compounds often have a common mechanism of action, as the target is the same.

15.1.2 Drug–target Interaction:

The biochemical processes such as metabolism (which is responsible for breaking down the food molecules and harvest energy in the form of ATP and biosynthesis of necessary biomolecules from the available precursor molecules using many enzymes),cell-signaling (senses any change in the environment using the receptor molecules and send signals to various processes to elicit an appropriate response) etc… are essential for the normal functioning of our body. These routine processes may be disturbed by any external factors such as microorganism, chemicals etc.. or by a disorder in the system itself. Under such conditions we may have to take medicines to restore the normal functioning of the body.

These drug molecules interact with biomolecules such as proteins, lipids, etc..that are responsible for different functions of the body. For example, proteins which act as biological catalysts are called enzymes and those which are important for communication systems are called receptors. The drug interacts with these molecules and modify the normal biochemical reactions either by modifying the enzyme activity or by stimulating/suppressing certain receptors.

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